

BTMACS201**B. Tech. / B. Tech. + M.B.A. / B. Tech. + M. Tech.**[IT/CSE/CCE/CSE-BDA/CSE-CF/CSE-CC/BDCE-Impetus/CSE-ICS/CSE-MA/CSE-BDAI/
CSE-CMCI/CSE-ES/CSE-DSI/CSE-FSDI/CSE-AII/CSE-AIMLM]**II Semester Examination, July - August 2022****Mathematics - II**

Choice Based Credit System (CBCS)

Duration: 3 Hrs.**Maximum Marks: 60****Minimum Pass Marks: 24**

- Note:
1. All questions carry equal marks, out of which part 'A' and 'B' carry 3 marks and part 'C' carries 6 marks.
 2. From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice.
 3. Draw neat diagram, wherever necessary.
 4. Assume suitable data wherever necessary.
 5. Symbols and notations used have their usual meaning.

Q.1(A) If λ is an eigen value of A , then λ^m is an eigen value of A^m .**03****(B)** Prove that similar matrices have same eigenvalues.**03****(C)** Reduce the following matrix into normal form and find its rank:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

06**OR**

Show that the following system is consistent and solve it

$$x + 2y - 5z = -9, 3x - y + 2z = 5, 2x + 3y - z = 3$$

Q.2(A) Solve: $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$ **03****(B)** Solve: $\frac{dy}{dx} + \frac{y}{x} = x^2$, given $y = 1$ when $x = 1$.**03****(C)** Solve by using method of variation of parameters $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = \frac{e^{3x}}{x^2}$.**06****OR**

$$\text{Solve: } \frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 2y = e^x + \cos x.$$

Q.3(A) Prove the following relations $\Delta + \nabla = \frac{\Delta}{\nabla} - \frac{\nabla}{\Delta}$.**03****(B)** Express $x^3 - 2x^2 + x - 1$ into factorial polynomial and hence show that $\Delta^4 y = 0$.**03****(C)** Use Lagrange's interpolation formula to find $f(4)$, where

x	0	2	3	6
$y = f(x)$	648	704	729	792

06**OR**Find $\frac{dy}{dx}$ at $x = 1.5$ from the following table

x	1.5	2.0	2.5	3.0	3.5	4.0
$y = f(x)$	3.375	7.0	13.625	24.0	38.875	59.0

Contd...

Q.4(A) Apply Euler's method and solve for $x = 0.6$ from $\frac{dy}{dx} = 1 - 2xy, y(0) = 0$, take $h = 0.2$ 03

(B) Explain Runge-Kutta method of fourth order to solve $\frac{dy}{dx} = f(x, y), y(x_0) = y_0$. 03

(C) The differential equation $\frac{dy}{dx} = 1 + y^2$ satisfies the following set of values of x and y

x	0	0.2	0.4	0.6
y	0	0.2027	0.4228	0.6841

Compute $y(0.8)$, using Milne's method. 06

OR

Apply Euler's modified method to compute y for $x = 0.4$ by taking $h = 0.1$.

Given $\frac{dy}{dx} = x + y, x_0=0, y_0=1$

Q.5(A) Define continuous and discrete random variable, with example. 03

(B) If 3% of the electric bulbs manufactured by a company are defective, find the probability that in a sample of 100 bulbs exactly five bulbs are defective. 03

(C) X is a normal variable with mean 30 and standard deviation 5, find the probability that $26 \leq X \leq 40$, (given that $p(0 \leq z \leq 0.8) = 0.2881$ and $p(0 \leq z \leq 2) = 0.4772$) 06

OR

Consider a system that can be in one of two possible states, $S = \{0,1\}$. In particular,

suppose that the transition matrix is given by $P = \begin{bmatrix} \frac{1}{2} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix}$ and the system is in state 0 at

time $n = 0$, i.e., $X_0 = 0$. Draw the state transition diagram and find the stationary distribution.

XXXXXXX